Question 1. Referencing managers (2 marks)

(a) What word processor and reference manager are you using for this assignment? (see also Question 8 where you need to demonstrate the references in your reference manager) (1 mark)

I am using Microsoft Word as word processor and Zotero as reference manager for this assignment.

(b) Will you use cite-as-you-write when preparing this assignment? If not, why not? (1 mark)

Yes, I will use cite-as-you-write function in Zotero when preparing this assignment. As it ensures formal IEEE formatting and accurate citations throughout this assignment.

Question 2. Open Access Journals (13 marks)

There are generally two types of platforms for publishing research outcomes: open access journals and non-open access (closed access) journals.

For this question, please conduct your own background research to identify and compare the features and differences between open access and non-open access (closed access) journals. Cite all your sources and use complete and correctly formatted references (in IEEE format, including the dates of access if required). Collate all your references in one list at the end of the assignment.

(a) What role does open access journal play in disseminating scholarly research? Cite your source(s). (<100 words, 1 mark)

Open access journals break down the paywall barriers that have long hindered the dissemination of scientific innovation and scholarly work. Open access journals deliver significant global impact, successfully attracting diverse research topics, authors, and readers from around the world. As concluded by Saqr et al., open access is a feasible and rewarding model that helps researchers in the Global South disseminate their scholarly work, which is often neglected by mainstream commercial and established publications.

(b) What are the key differences between publishing research papers in open access journals and in non-open access (closed access) journals? Cite your source(s). (<150 words, 1 mark)

Closed access journals lock published works behind paywalls, restricting access to a limited audience of subscribers. In contrast, open access (OA) journals break down these barriers, offering unrestricted access so anyone worldwide can read papers. OA empowers local journals to help Global South researchers disseminate their work, which is often sidelined by closed access publications. Financially, OA typically charges authors (via APCs) rather than readers, while closed access shifts costs to readers or institutions instead of authors[1].

(c) What are the main advantages and disadvantages of publishing research papers in open access journals? Cite your source(s). (<150 words, 2 marks)

Open access journals break down paywalls, allowing anyone worldwide to access academic publications freely. They also help researchers in the Global South disseminate their work more widely. However, a key disadvantage is that authors are required to pay article processing charges (APCs), which can create a significant financial burden[1]

(d) Each published paper is labeled with a digital object identifier (F). For example, 10.1109/ACCESS.2020.2988796 is the DOI of a paper published in IEEE Access. Find the detailed information of this paper (e.g., title, authors, publication year, etc), provide a proper citation, and include it in your reference list. (<100 words, 1 mark)

The paper with DOI 10.1109/ACCESS.2020.2988796 is titled Unsupervised K-Means Clustering Algorithm, authored by Kristina P. Sinaga and Miin-Shen Yang. It was published in IEEE Access in 2020, with volume 8 and pages 80716–80727

[2] K. P. Sinaga and M.-S. Yang, "Unsupervised K-Means Clustering Algorithm," IEEE Access, vol. 8, pp. 80716–80727, 2020, doi: 10.1109/ACCESS.2020.2988796.

The following question parts (e) – (h) refer to the article: **Rappaport, S. T., et al. Millimeter Wave Mobile Communications for 5G Cellular: It Will Work! Ieee Access**

(e) Find the paper and obtain its pdf. Enter the article in your reference manager. Note that the reference above is incomplete and not in a standard format. Ensure that you identify any missing details and correctly format the article in your reference manager according to the IEEE referencing style.

Provide an in-text citation here and ensure that the full citation is correctly formatted in the reference list at the end of the assignment. (<50 words, 1 mark)

This paper investigates the potential of millimeter-wave communications for 5G cellular systems. [3]

[3] T. S. Rappaport et al., "Millimeter Wave Mobile Communications for 5G Cellular: It Will Work!," IEEE Access, vol. 1, pp. 335–349, 2013, doi: 10.1109/ACCESS.2013.2260813.

(f) In your own words, summarise what the paper is about in less than 50 words, based on the first two lines of the paper abstract. Use appropriate paraphrasing techniques, and specify which paraphrasing methods you have used from the following: (3 marks)

Wireless operators' worldwide bandwidth scarcity has driven investigation into underexploited mm-wave spectrum for 5G networks, though urban propagation properties remain poorly understood in dense environments[3].

Paraphrasing methods used: Synonyms; Changing word order; Changing conjunctions / linkers

☑ Synonyms

☑ Changing voice e.g. passive to active OR changing word order

□ Changing a clause to a phrase

□ Changing parts of speech
☑ Changing conjunctions / linkers

(g) Fill in the table below, reporting exactly how many times the article by Rappaport et al. has been cited on Google Scholar and Web of Science on the date(s) you accessed them. (2 marks)

Article version	Number of Google Scholar citations	Number of Web of Science citations	Date(s) of access
Peer-reviewed	9445	5540	10/04/2026

(h) For this question you need to find the list of papers that cited Rappaport et al. ordered using the "Sort by relevance" option on the left hand side of Google Scholar. Identify the publishing platform (i.e., open access, non-open access, or hybrid) of the 8 most relevant papers that cite the article by Rappaport et al. Fill in the table below with the number of citing papers (out of the 8 most relevant citing papers) that were published in open access, non-open access, or hybrid journals.
Hint: A hybrid journal allows both open access and non-open access modes.
Using the information in your table, in one sentence, summarise the types of publishing platform that the first 8 references to Rappaport et al. in Google Scholar are found.
(<100 words, 2 marks)

	Open Access	Non-Open Access	Hybrid	Date of access
Number of citing papers	3	0	5	10/04/2026

In Google Scholar, the 8 most relevant citing papers to Rappaport et al. include 3 open access and 5 hybrid publications, with zero non-open access papers. This indicates a dominance of open access and hybrid publishing models in the most relevant references.

Question 3. Referencing and Plagiarism. **(6 marks)**
Below is a passage in an (imaginary) deliverable that a student has submitted to answer the question: "Can Deep Neural Networks Learn Physics Laws?", and further below are excerpts from one of the original sources they used.

Student's answer:
Deep learning techniques are feasible to physics data. For example, measured physics data can be directly used as the input of a neural network to perform a pre-processing (Li et al., 2021). In addition, deep neural networks can be trained to generate images based on the measured physics data (Li et al., 2021). In these approaches, the physics of measurement is implicitly embedded in the network, and the computation can be accelerated with the help of masive parallelization (Li et al., 2021).

Original source:
Li, M., Guo, R., Zhang, K., Lin, Z., Yang, F., Xu, S., Chen, X., Massa, A., & Abubakar, A.

(2021). Machine learning in electromagnetics with applications to biomedical imaging: A review. *IEEE Antennas and Propagation Magazine, 63* (3), 39–51.

Deep learning techniques can be applied to the imaging process in several stages. They can be directly applied to preprocess measured data [14]–[16]. For instance, filters for noise removal or signal classification can be constructed with a trained network based on previous data. In the imaging process, end-to end DNNs can be trained to transform the measured data directly to images [17], [18]. In these approaches, the physics of measurement is implicitly embedded in the network, and the computation can be accelerated with the help of massive parallelization.

Answer the following two questions about this:

(a) Identify any problems with non-originality, plagiarism and/or referencing style in the student's answer. List the problems, being very specific about where the problems are. (4 marks)

Non-originality

Location: Most of the paragraph of the student's answer.

Issue: The paragraph only presents the author's ideas with no student voice or original argument. This violates academic writing principles that require the student's own ideas to be central, with sources used only for support.

Plagiarism

Location: The last sentence and the phrase "deep neural networks can be trained to generate images based on the measured physics data".

Issue: The last sentence is copied verbatim from the source. The other phrase only changes a few words (e.g., DNNs to deep neural networks). Although the student cites (Li et al., 2021), they do not use quotation marks for directly copied text.

Referencing Style

Location 1: The end of every sentence.

Issue: The student cites (Li et al., 2021) after every sentence unnecessarily. Only one citation at the end of the paragraph is required.

Location 2: The last sentence.

Issue: The sentence is copied directly from the source without quotation marks, which is improper citation.

Location 3: The term "masive parallelization".

Issue: There is a spelling mistake; the correct spelling is "massive parallelization".

(b) State, using examples, what the student should have done to avoid the problems. (2 marks)

- **Fix non-originality**

Action: Open the paragraph with the student's own voice, including a clear topic sentence and main argument, rather than only restating source content.

Example:

Deep neural networks are capable of learning physical laws through processing physics data, and have practical applications in imaging workflows.

- **Fix plagiarism**

Action: Enclose verbatim copied text (e.g., the final sentence) in quotation marks, and fully paraphrase the clause “DNNs can be trained to transform the measured data directly to images” instead of making only minor word changes.
Example:
"the physics of measurement is implicitly embedded in the network, and the computation can be accelerated with the help of massive parallelization" (Li et al., 2021).
The original phrase "DNNs can be trained to transform the measured data directly to images" can be paraphrased to "deep neural networks can be used to reconstruct images from raw measured data".

- **Fix referencing style**

Action: Remove redundant repeated citations, correct spelling errors, and properly format direct quotations with quotation marks.
Example:
Delete the repeated "(Li et al., 2021)" citations after every sentence, and place a single citation at the end of the paragraph. Correct the spelling error "masive parallelization" to "massive parallelization", and enclose the directly copied final sentence in quotation marks.

Question 4. Quality metrics for journals and journal articles (5 marks)

Question 4 and Question 5 refer to the two articles cited below. When answering questions, remember to include the databases you used and the dates of access, e.g., h-index = 42 from [database] accessed on [date].

Article 1

Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., Cui, C., Corrado, G., Thrun, S., & Dean, J. (2019). A guide to deep learning in healthcare. *Nature medicine*, *25*(1), 24– 29. doi: https://doi.org/10.1038/s41591-018-0316-z.

Article 2

E. J. Bond, X. Li, S. C. Hagness, and B. D. Van Veen, “Microwave imaging via space-time beamforming for early detection of breast cancer,” *IEEE Trans. Antennas Propag.*, vol. 51, no. 8, pp. 1690–1705, Aug. 2003, doi: https://doi.org/10.1109/TAP.2003.815446.

(a) The two citations above are in different formats, APA and IEEE style respectively. What are key differences between these two formats in terms of their in-text citations, and the ordering of the reference list? Show an example of an in-text citation for Article 1 using IEEE style and APA style formatting. (2 marks)

Key differences in terms of in-text citations

- APA style: Uses the author-date format (e.g., (Esteva et al., 2019)), which includes the first author’s last name and the publication year. For works with 3 or more authors, the citation is simplified to the first author’s last name followed by et al. (meaning "and others").
- IEEE style: Uses the numbered format (e.g., [1]), where the bracketed Arabic numeral corresponds directly to the entry order in the reference list.

Key differences in terms of reference list

- APA reference list: Entries are sorted alphabetically by the first author’s last name, it not related to the first citation.

- IEEE reference list: Entries are sorted by the order of their first citation in the work, with numbers corresponds to the in-text citations.

- The example of an in-text citation for Article 1 using IEEE style. [7]
  [7] A. Esteva et al., “A guide to deep learning in healthcare,” Nat. Med., vol. 25, no. 1, pp. 24–29, Jan. 2019, doi: 10.1038/s41591-018-0316-z.
- The example of an in-text citation for Article 1 using APA style. (Esteva et al., 2019)
  Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., Cui, C., Corrado, G., Thrun, S., & Dean, J. (2019). A guide to deep learning in healthcare. Nature Medicine, 25(1), 24–29. https://doi.org/10.1038/s41591-018-0316-z

(b) Determine and report the h-Index of the last author in Article 2 based on Scopus, Google Scholar and WoS. Is the h-Index of the author the same or different across these three databases? Give a suitable explanation for your answer. (2 marks)
The h-index values are different across the three databases.
- h-index = 41 from Scopus accessed on 10/04/2026.
- h-index = 58 from Google Scholar accessed on 10/04/2026.
- h-index = 39 from WoS accessed on 10/04/2026.

- Google Scholar: It offers the broadest coverage, encompassing journal articles, conference papers, preprints, and technical reports. This extensive scope results in the highest h-index for the author.
- Scopus: It indexes a curated selection of peer-reviewed content but excludes non-peer-reviewed works. Consequently, its h-index is typically lower than that reported in Google Scholar.
- Web of Science (WoS): It employs the strictest inclusion criteria, primarily prioritizing high-impact, peer-reviewed journals. This focus on elite publications leads to the lowest h-index among the three databases.
- Additionally: Variations exist across databases in citation counting rules, which further lead to the differences in reported h-index values.

(c) Choose one metric suitable for assessing the quality or impact of Article 2. Justify your choice and report its value. (1 mark)
Selected Metric: **Impact Factor (IF)**
IF is a standard metric for evaluating journal quality. Article 2 was published in *IEEE Transactions*, a top-tier journal. The IF is calculated as the average number of citations received per article published in the journal over the last two years. The 2024 Journal Impact Factor for this journal is 5.8 from Web of Science accessed on 12/04/2026.

Question 5. Teams of authors (3 marks)
The authorship list can reveal important information about a paper. Fill in the table below answering the following questions for the published version of Article 2:
i) Who is the corresponding author? (Write N/A if not explicitly stated in the paper.)
The corresponding author is E.J. Bond, who is listed as a PhD student

ii) What were the academic or industry affiliations of the authors at the time of publication?

iii) What were their positions (e.g. professor, MPhil student, PhD student)?

Author order	Name	Affiliation	Academic or professional position	Correspon ding author?
Author #1	E. J. Bond	University of Wisconsin-Madison	PhD Student	**Yes**
Author #2	X. Li	University of Wisconsin-Madison	PhD Student	No
Author #3	S. C. Hagness	University of Wisconsin-Madison	Associate Professor	No
Author #4	B. D. Van Veen	University of Wisconsin-Madison	Professor (Principal Investigator)	No

Question 6. Research Ethics (6 marks)

(a) Imagine you are collaborating with a company on the development of a clinical microwave brain imaging system. Discuss the ethical considerations surrounding data collection, ownership, confidentiality, and sharing that must be addressed. (4 marks)

This answer is based on REIT6811 Lecture 5: Research with Humans and Lecture 2: Academic and research Integrity

In terms of data collection, researchers must obtain ethics approval by **The University of Queensland Human Research Ethics Committee (HREC)** and get consent from participants before conducting any study involving human participants, and only collect necessary data.(Lecture 5, slide 46-49) Regarding ownership, it is important to clarify data ownership before the project and access rights between the relevant groups. Additionally, no organization or individual can keep or share without permission. A data management plan should be established to define rights and use. All data should be stored securely to prevent misuse or unauthorized access. All actions are taken to protect the participants' privacy.(Lecture 5, slide 46, Lecture 2, slide 20) With respect to confidentiality, de-identify some data, including removing or blurring facial features in scans, to protect privacy. (Lecture 5, slide 46 47) Finally, sharing requires obtaining explicit consent for data sharing, share only de-identified data, and inform participants that full withdrawal may not be possible after publication. (Lecture 5, slide 47 48)[10], [11]

(b) When evaluating the collected clinical data, you find that some of the data points don't fit the expected trend, and you get significantly better performance by removing those data points. Can these data points be removed for publication, and why or why not? What are appropriate steps to do in this situation? (2 marks)

These data points cannot be removed for publication. "Never alter your data!"(Lecture 2, slide 20)

Reason:

Removing data points because it doesn't fit the expected trend constitutes research misconduct and violates core principles of academic integrity.

appropriate steps

- Investigate the root cause of the outliers: Verify whether the data points are valid, or if they result from verifiable errors such as equipment malfunction, measurement mistakes, or data entry issues.
- Handle the data transparently in line with academic integrity: If the data is confirmed to be invalid (e.g., due to equipment failure), it may be excluded, but the reasons for removal must be fully documented and transparently reported in the work. If the data is valid, it cannot be removed simply for not fitting the expected trend; instead, its impact on the study results must be fully discussed in the paper.[10]

Question 7. Conflicts of Interest (2 marks)

Dr. X is a member of a national research grant committee panel. Recently, Dr. X received a request to assess a grant project led by Dr. Y, who was Dr. X's classmate during their university studies. Dr. X had a close relationship with Dr. Y at that time, and they have maintained regular personal contact since graduation. Despite this relationship, Dr. X decides to accept the assessment task from the panel.

(a) Identify the conflict of interest (COI) that may arise from Dr. X's grant assessment, as well as the potential ethical concerns associated with this task.

(<100 words, 1 mark)

This is a Personal Relationships – family/friends COI. Dr. X had a close relationship with Dr. Y at that time, which may lead to unfairness. the potential ethical concerns associated with this task are unfair assessment, loss of honesty, respect, accountability of ethic principles.

(b) How can Dr. X manage this conflict of interest to maintain research integrity?

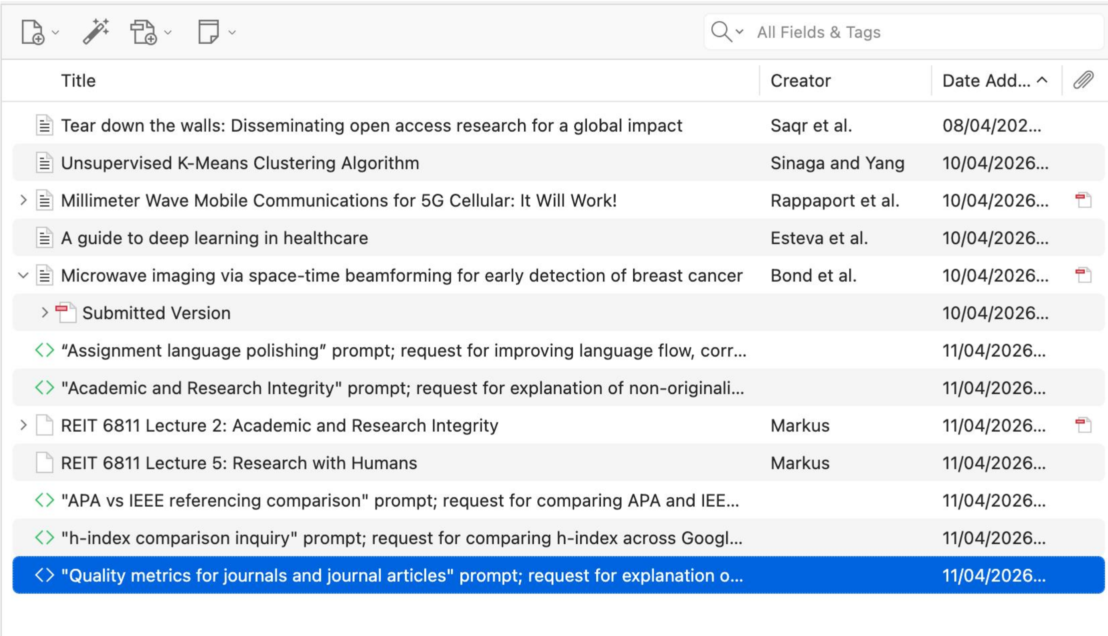
(<100 words, 1 mark)

Dr. X should disclose and register the COI formally and immediately and remove themselves from their involvement in the assessment to ensure fairness and research integrity.[10]

Question 8. Demonstrating your use of a reference manager (3 marks)

Add a Reference section in IEEE format which includes all the references in your assignment. Note that you need to cite answers using concepts or text provided by generative AI just as you would any other source (see Appendix and table A1 for details of the format requirements).

Add an Appendix to demonstrate the list of articles in your reference manager using either a screen image ("screenshot"), or a copy of your Bibtex file.



[1] M. Saqr, A. Al-Mohaimeed, and Z. Rasheed, "Tear down the walls: Disseminating open access research for a global impact," *Int. J. Health Sci.*, vol. 14, no. 5, pp. 43–49, Oct. 2020.

[2] K. P. Sinaga and M.-S. Yang, "Unsupervised K-Means Clustering Algorithm," *IEEE Access*, vol. 8, pp. 80716–80727, 2020, doi: 10.1109/ACCESS.2020.2988796.

[3] T. S. Rappaport *et al.*, "Millimeter Wave Mobile Communications for 5G Cellular: It Will Work!," *IEEE Access*, vol. 1, pp. 335–349, 2013
doi: 10.1109/ACCESS.2013.2260813.

[4] *"Assignment language polishing" prompt; request for improving language flow, correct grammar, and refine sentence structure*, Doubao, ByteDance, 2026, Accessed: Apr. 08, 2026. [Online]. Available: https://www.doubao.com.

[5] *"Academic and Research Integrity" prompt; request for explanation of non-originality, plagiarism, referencing style*, Doubao, ByteDance, 2026, Accessed: Apr. 11, 2026. [Online]. Available: https://www.doubao.com.

[6] *"APA vs IEEE referencing comparison" prompt; request for comparing APA and IEEE referencing styles*, Doubao, ByteDance, 2026, Accessed: Apr. 11, 2026. [Online]. Available: https://www.doubao.com.

[7] A. Esteva *et al.*, "A guide to deep learning in healthcare," *Nat. Med.*, vol. 25, no. 1, pp. 24–29, Jan. 2019, doi: 10.1038/s41591-018-0316-z.

[8] *"h-index comparison inquiry" prompt; request for comparing h-index and any organization or individual across Google Scholar, WoS, and Scopus*, Doubao, ByteDance, 2026, Accessed: Apr. 11, 2026. [Online]. Available: https://www.doubao.com.

[9] *“Quality metrics for journals and journal articles” prompt; request for explanation of Impact Factor*, Doubao, ByteDance, 2026, Accessed: Apr. 11, 2026. [Online]. Available: https://www.doubao.com.
[10]B. Markus, “REIT 6811 Lecture 2: Academic and Research Integrity.” Accessed: Apr. 11, 2026, [Online]. Available: https://learn.uq.edu.au/ultra/courses/_198751_1/outline/edit/document/_12806479_1?courseId=_198751_1&view=content&state=view.
[11]B. Markus, “REIT 6811 Lecture 5: Research with Humans.” Accessed: Apr. 11, 2026. [Online]. Available: https://learn.uq.edu.au/ultra/courses/_198751_1/outline/edit/document/_12806479_1?courseId=_198751_1&view=content&state=view.

I acknowledge the use of generative AI tools in completing this assignment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content and ensuring it adheres to academic integrity standards.

Table A1. Generative AI usage (tick all that apply)

AI Model Used and date	**Table A1. Generative AI usage** (tick all that apply)								
	Language Translation	Grammar/Style	Planning/Drafting	Research/ Background Information	Content Creation text	Content Creation visual	Content Creation code	Feedback	Other (provide details)
Doubao *08/04/26*	√	√						√	*Assignment language polishing & grammar correction*
Doubao *11/04/26*	√	√	√	√					*Assignment planning, research supplement & draft writing*